

MANISH CHHETRI

Phone: (+977) 9814171851 , Email: hsinam12man34@gmail.com

Website: manishchhetri.com.np , Google scholar: [manish](#)

Github: [chhetri123](#) LinkedIn: [chhetri123](#)

PROFILE

Results-driven and passionate professional with strong research interest in AI and Cloud Computing. Extensive experience in software development, cloud services, and mentoring junior developers. Skilled in Docker, Kubernetes, and microservices, with a focus on secure, scalable solutions.

Interests: Artificial Intelligence, Cloud Computing, Machine Learning, NLP, Computer Vision, IoT.

EDUCATION

Western Region Campus, Institute of Engineering, TU, Nepal

2019 – 2024

B.E. in Electronics, Communication and Information Engineering

Final Grade: First Division

Percentage: 72.55%

Thesis Title:

Image Captioning in Nepali Language

Global Collegiate School, Nepal

Science, National Examination Board (NEB)

2017–2019

Final Grade: First Division

CGPA: 3.45/4.0

SKILLS AND COMPETENCIES

Languages:	Python, C/C++, JavaScript (Node.js), Arduino
Machine Learning:	Transformers, LLMs, CNNs, RNNs
Web Development:	Node.js, TypeScript, Express, NestJS, React.js, Next.js
Database:	PostgreSQL, MySQL, MongoDB
DevOps & Cloud:	Docker, Kubernetes, CI/CD (GitHub Actions), AWS (EC2, S3, Lambda, VPC)
Hardware:	Arduino, SIMmodule, GPS, Bluetooth, VHDL (Embedded Programming)
Strengths:	Communication, Team Leadership, Time Management, Project Coordination

PUBLICATIONS

Nepali Image Captioning: Generating Coherent Paragraph-Length Descriptions Using Transformer

Journal of Soft Computing Paradigm,
6(1), 70-84

CO-AUTHOR

2024

- Subedi, N., Paudel, N., Chhetri, M., Acharya, S., & Lamichhane, N
- DOI: [10.36548/jscp.2024.1.006](https://doi.org/10.36548/jscp.2024.1.006)

Drowsiness and Crash Detection Mobile Application for Vehicle's Safety

Journal of IoT in Social, Mobile, Analytics, and Cloud, 6(1), 54-66

CO-AUTHOR

2024

- Subedi, N., Paudel, N., Chhetri, M., Acharya, S., & Lamichhane, N.
- DOI: [10.36548/jismac.2024.1.005](https://doi.org/10.36548/jismac.2024.1.005)

RESEARCH EXPERIENCE

Nepali Image Captioning

2024

- Created Nepali Paragraph Captioning Dataset using BeautifulSoup for web scraping and manual annotation.
- Developed a web app using Node.js and React, deployed models to Hugging Face for inferencing.
- Built image captioning models using ResNet50, ResNet152, InceptionV3 for feature extraction and LSTM/Transformer for Nepali caption generation.
- Authored a comprehensive scientific paper and presented at an international conference, receiving recognition for its impact.
- [Using LSTM](#)
- [Using Transformer](#)

Crash and Drowsiness Detection System

2023

- Integrated Arduino, GPS, GSM, Accelerometer, and Bluetooth module to detect collisions and notify rescue centers.
- Implemented false notification prevention using a mobile app connected to the crash system.
- Implemented facial detection system to monitor driver drowsiness.
- Authored and submitted a manuscript, serving as corresponding author.
- [CODE](#)

CONFERENCE/WORKSHOP ATTENDED

Recent Trends on Science, Technology and Innovation 2024 (RTSTI) *Institute of Forestry, Pokhara Campus*
PRESENTER 2024

- Presented a comprehensive research on Vehicle's Crash Detection and Alert System.
- Received positive feedback for innovative work on combining hardware with AI.

INDUSTRIAL EXPERIENCE

Maitri Services

Software Engineer

Lalitpur, Kathmandu Nepal

Oct. 2025 – Present

- Designing backend architecture and database schemas with a focus on clean code structure, modularization, and long-term maintainability.
- Building dynamic, schema-driven data systems using JSON Schema, enabling frontend UI and workflow rendering from backend configurations.
- Developing agents and sub-agents for batch-based processing of large documents (100+ pages), structured content extraction, and automated workflow triggering and management.
- Developing and maintaining 25+ cloud-based services and processing pipelines on GCP, while collaborating with engineers on system design, performance optimization, CI/CD improvements, and architectural trade-offs.
- Actively collaborating with teams and senior engineers on system design discussions, performance optimization, and architectural trade-off analysis.

Swivt Technology
Full Stack Software Engineer

Lalitpur, Kathmandu Nepal
Nov. 2023 – Aug. 2025

- Managed server configurations and resolved production issues by analyzing logs, reducing downtime by ~30%
- Delivered 5+ international client projects with high-quality results and consistently positive feedback.
- Added key features to existing systems, including Payment Systems, Admin Panel, CMS, CRM, WebRTC, and role-based access control.
- Containerized applications, optimized Nginx, managed Cloudflare DNS/domains, and deployed RAG-based transformer models for a medical insurance system in a cloud-native setup.

EnterLeaf
Microservice Backend Developer

Kuleshwor, Kathmandu Nepal
Jul. 2023 - Nov. 2023

- Designed and developed different services using NestJS, leveraging NATS for message brokering, enabling a robust event-driven architecture.
- Integrated an API Gateway to handle and route requests efficiently from both mobile and web browser clients, ensuring seamless communication across services
- Implemented a CI/CD pipeline using GitHub Actions, streamlining the deployment process and ensuring continuous integration and delivery for rapid and reliable updates.

UpStartPrep
Backend Developer

Knoxville, Tennessee, United States
Dec. 2022 -Jun. 2023

- Creating and improving application programming interfaces (APIs), architecture, and servers, and databases (SQL) for E-Learning Platforms.
- Implementation of Payment System (Stripe) and Used Cloud Services like AWS.
- Implementation of microservices-based architecture, incorporating containerization and CI/CD pipeline for efficient development.

AWARDS AND HONOURS

Robo Soccer Competition
First Position

National Innovation Center Nepal, Pokhara
Jan. 2019

- Designed and developed an autonomous robot in a team of four that could play soccer, kick the ball, and defend the goal.

Battle for Speed
Second Position

VECTOR – Second National Technical Festival, WRC
Jan. 2019

- Developed an automatic line-following robot using Arduino programming, focusing on speed and precision.

Datathon Competition
First Position

I-CES, IOE Western Region Campus
Jan. 2019

- Collaborated in a two-member team to preprocess data, train a machine learning model with 91% accuracy, and present the results successfully.